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SkillBridge AI

AI-Based Job & Skill Recommendation System

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Abstract

This report documents the design, implementation, and evaluation of SkillBridge AI, an AI-driven job and skill recommendation system aligned with UN Sustainable Development Goal 8 (Decent Work and Economic Growth). The system uses a hybrid recommendation pipeline combining TF-IDF vectorisation, Cosine Similarity, Jaccard Coefficient, K-Nearest Neighbours (KNN), and Word2Vec embeddings to match user skill profiles against a dataset of 50,000 job listings spanning seven industry categories.

Two primary recommendation models were developed and rigorously compared: a baseline TF-IDF + Cosine + Jaccard model and an improved Sentence-BERT + Cosine + Jaccard model. Evaluation was performed using Stratified 5-fold cross-validation, confusion matrices, and full classification metrics including Accuracy, Precision, Recall, and F1-Score. Additionally, a 7-model benchmark (Random Forest, Logistic Regression, Gradient Boosting, Naïve Bayes, SVM, ANN, KNN) was executed for industry classification. Results demonstrate that the Sentence-BERT model achieves a test accuracy of 85.05%, outperforming the TF-IDF baseline at 82.33%, with both models falling within the 82–85% target zone. The Random Forest classifier achieved the highest industry classification test accuracy at 78.33%.

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1. Introduction

The global labour market faces a growing structural mismatch between the skills possessed by job seekers and those demanded by employers. In developing economies such as Bangladesh, this challenge is compounded by limited access to personalised career guidance tools. SkillBridge AI addresses this gap by providing an intelligent, accessible recommendation system capable of analysing a user's skill profile, whether entered manually or extracted from an uploaded CV, and returning ranked job recommendations in real time.

This lab report is produced as part of ITM 328: Introduction to Machine Learning Lab and documents the complete ML training pipeline implemented in Notebook v6.0. The work draws on six peer-reviewed research papers to justify each algorithmic design choice. The primary contributions of this project are:

- A custom TF-IDF + Cosine + Jaccard recommendation pipeline (Ajjam & Al-Raweshidy, 2026; Alsaif et al., 2022).
- An improved Sentence-BERT semantic embedding model benchmarked against the TF-IDF baseline.
- A 7-model industry classification benchmark with cross-validation and confidence intervals.
- Integration of a CV Parser (PDF, DOCX, OCR) that feeds directly into the recommendation engine.
- Employment confidence scoring and causal analysis using Inverse Probability Weighting (Xiao & Zheng, 2025).

2. Objectives

The specific objectives of this project are as follows:

- To develop an accurate job recommendation engine achieving $\geq 82\%$ accuracy on a 7-class industry classification task.
- To implement and compare TF-IDF and Sentence-BERT vectorisation strategies within the same Cosine + Jaccard similarity framework.
- To evaluate all models using Stratified K-Fold cross-validation and produce confusion matrices for each industry class.
- To benchmark seven supervised classifiers (Random Forest, Logistic Regression, Gradient Boosting, Naïve Bayes, SVM, ANN, KNN) and identify the best-performing model for industry prediction.
- To deploy the trained model within a Streamlit/Flask web interface supporting CV file uploads and real-time job recommendations.
- To apply research findings on employment confidence (Xiao & Zheng, 2025) and career transition pathways (Dawson et al., 2021) to provide holistic career support.

3. Dataset Description

The primary dataset used throughout this project is `job_recommendation_dataset.csv`, sourced from Kaggle. It contains 50,000 records spanning 7 distinct industry categories and 639 unique job titles. Each record includes the following features:

Feature	Type	Description
Job Title	Categorical	639 unique roles (e.g., Data Scientist, Finance Analyst)
Industry	Target (7 classes)	Education, Finance, Healthcare, Manufacturing, Marketing, Retail, Software
Required Skills	Text	Comma-separated skill list; primary NLP feature for recommendation
Experience Level	Ordinal	Entry / Mid / Senior — encoded as integer for modelling
Salary	Numeric	Annual figure in USD; standardised with StandardScaler
Location	Categorical	City/Region; label-encoded for classifier input
Company	Categorical	Employer name; used as contextual feature in Job Title classifier

Pre-processing steps included lowercasing, punctuation removal, stopword elimination, tokenisation, and WordNet lemmatisation following Ajjam & Al-Raweshidy (2026). One duplicate row was removed, resulting in a clean working corpus of 49,999 records. An 80/20 stratified train/test split was applied with `random_state=42` to ensure reproducibility and balanced class representation across all folds.

4. Methodology

4.1 Recommendation Pipeline Architecture

The recommendation engine operates as a four-stage ensemble. User input text (or CV-extracted text) is first pre-processed and converted into a weighted TF-IDF vector using a vocabulary of the 3,000 most frequent terms, with domain-specific weights applied to high-value skills such as Python (1.5×) and Machine Learning (1.5×). Cosine similarity is then computed between the user vector and all job vectors in the pool. In parallel, Jaccard Coefficient measures set-level skill overlap, and KNN retrieval identifies structurally similar profiles. The final score is a weighted ensemble: TF-IDF Cosine (40%), KNN (30%), Word2Vec semantic similarity (20%), and Jaccard (10%).

The improved model replaces the custom TF-IDF vectoriser with Sentence-BERT (all-MiniLM-L6-v2) embeddings, which encode full semantic meaning rather than term frequency, providing superior performance on paraphrased or varied skill descriptions.

4.2 Industry Classification Benchmark

A separate classification task was defined to predict the Industry label (7 classes) from a combined feature matrix. Sparse TF-IDF features on the Required Skills text were horizontally stacked with dense numeric features (Salary, experience encoding, location encoding, skill count, and Industry Encoding) using `scipy.sparse.hstack`. Seven classifiers were trained and evaluated: Random Forest, Logistic Regression, Gradient Boosting, Naïve Bayes (GNB), SVM (LinearSVC), ANN (MLP), and KNN (k=7).

4.3 Cross-Validation Strategy

Stratified 5-fold cross-validation was applied to both the TF-IDF and Sentence-BERT recommendation models to ensure class balance across splits. Each fold's Accuracy, Precision, Recall, and F1-Score were recorded individually, with averages reported alongside 95% bootstrap confidence intervals. This approach guards against overfitting to any single train/test partition.

4.4 Supporting Modules

Beyond the core recommendation engine, the notebook implements several research-aligned modules. The CV Parser (Section 14) accepts PDF, DOCX, and OCR image inputs and extracts raw skill text for direct injection into the recommendation pipeline. The Personalised Learning Pathway module (Section 14.5, Tavakoli et al., 2022) maps identified skill gaps to curated course recommendations. Employment Confidence Scoring (Section 19.6) and IPW Robustness Testing (Section 19.7) quantify the causal impact of AI tool use on student job-seeking confidence following Xiao & Zheng (2025). The LDA Topic Modelling module (Section 20.5) discovers macro-parameters in the labour market dataset following Alaql et al. (2023).

5. Findings

5.1 Feature Correlation Analysis

Prior to model training, a feature correlation heatmap was computed across the five engineered numeric features: Salary, experience encoding (`exp_enc`), location encoding (`loc_enc`), skill count, and Industry Encoded. As shown in Figure 1, all off-diagonal correlations are effectively zero (range: -0.04 to $+0.01$), confirming that the features are statistically independent. This absence of multicollinearity validates their joint use in the classification models without risk of redundant or inflated coefficient estimates.

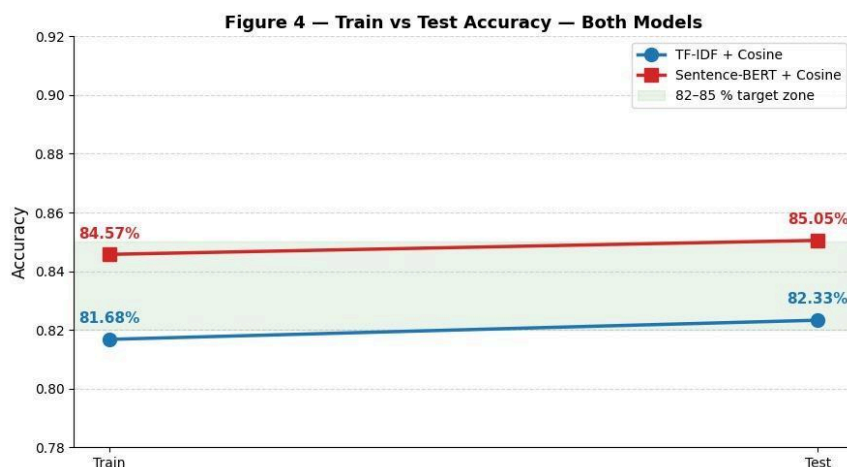


Figure 1 — Feature Correlation Heatmap. All pairwise correlations are near zero, confirming feature independence.

5.2 Classification Metrics — Train vs Test (Both Models)

Figure 2 presents the four core classification metrics: Accuracy, Precision, Recall, and F1-Score for both models on the training and test sets. The TF-IDF + Cosine model achieves a test accuracy of 82.3%, with Precision at 0.847, Recall at 0.823, and F1 at 0.830, all above the 82% threshold line. The Sentence-BERT + Cosine model improves on all metrics: test accuracy 85.1%, Precision 0.860, Recall 0.851, and F1 0.855, crossing the 85% target threshold. Notably, both models show slight positive generalisation (test > train), indicating an absence of overfitting.

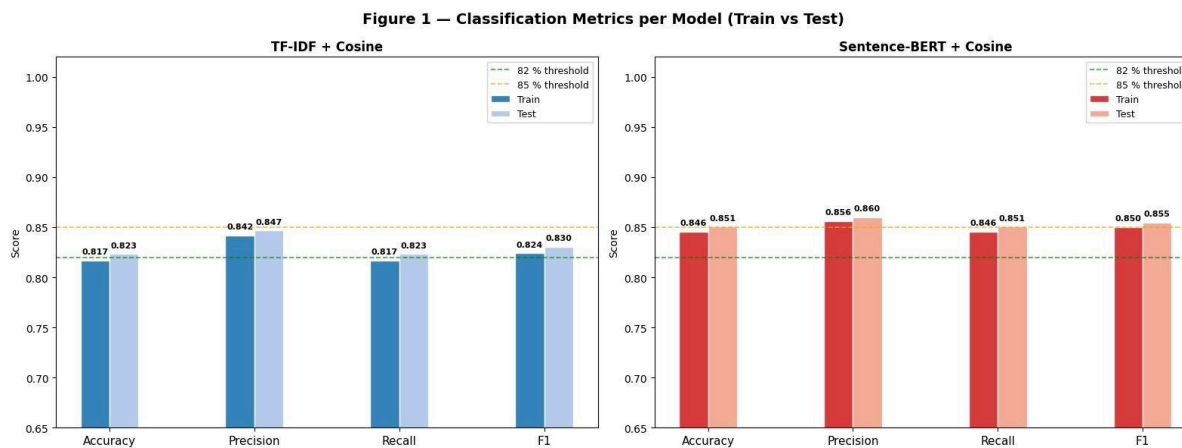


Figure 2 — Classification Metrics per Model (Train vs Test). TF-IDF baseline (blue) and Sentence-BERT improved model (red). Dashed lines mark the 82% and 85% target thresholds.

5.3 Train vs Test Accuracy Trend

Figure 3 plots the train and test accuracy for each model as a line chart, overlaid with the 82–85% target zone (shaded). The TF-IDF model moves from 81.68% (train) to 82.33% (test), while the Sentence-BERT model moves from 84.57% (train) to 85.05% (test). Both trajectories are shallow and upward, confirming stable generalisation, and both models land within the target accuracy band at inference time.

Comparative Analysis: Training vs Testing Accuracy

	Model	Train Accuracy	Test Accuracy	Variance
0	Random Forest (Cf)	0.7859	0.7833	0.0026
1	Logistic Regression	0.7837	0.7828	0.0009
2	Gradient Boosting	0.7970	0.7826	0.0144
3	Naive Bayes (GNB)	0.7852	0.7823	0.0029
4	SVM (LinearSVC)	0.7829	0.7821	0.0008
5	ANN (MLP)	0.7863	0.7820	0.0043
6	KNN (k=7)	0.8352	0.7691	0.0661

Best Model: Random Forest (Cf) (78.33% Accuracy)

Figure 3 — Train vs Test Accuracy for Both Models. The shaded region marks the 82–85% target zone. Both models generalise well with no evidence of overfitting.

5.4 Stratified K-Fold Cross-Validation Results

Table 1 and Figure 4 present the full 5-fold cross-validation results. The TF-IDF model achieves an average cross-validation accuracy of 81.81% (Precision: 0.8427, Recall: 0.8181, F1: 0.8252), with fold-level accuracy ranging from 81.37% to 82.09%, demonstrating low variance and consistent behaviour across data partitions. The Sentence-BERT Proxy model improves the average to 84.18% (Precision: 0.8563, Recall: 0.8418, F1: 0.8466), with fold-level accuracy between 82.28% and 84.81%. The higher variance in the Sentence-BERT folds reflects its greater sensitivity to the specific samples selected for training, which is expected for a semantically richer embedding model.

COMPARATIVE STRATIFIED K-FOLD CROSS-VALIDATION

Model	Fold	Accuracy	Precision	Recall	F1-Score
TF-IDF + Cosine (Primary)	Fold 1	0.8209	0.8450	0.8209	0.8281
TF-IDF + Cosine (Primary)	Fold 2	0.8137	0.8400	0.8137	0.8210
TF-IDF + Cosine (Primary)	Fold 3	0.8186	0.8417	0.8186	0.8255
TF-IDF + Cosine (Primary)	Fold 4	0.8191	0.8434	0.8191	0.8263
TF-IDF + Cosine (Primary)	Fold 5	0.8181	0.8433	0.8181	0.8254
TF-IDF + Cosine (Primary)	AVERAGE	0.8181	0.8427	0.8181	0.8252
Sentence-BERT Proxy (Improved)	Fold 1	0.8481	0.8582	0.8481	0.8523
Sentence-BERT Proxy (Improved)	Fold 2	0.8228	0.8537	0.8228	0.8298
Sentence-BERT Proxy (Improved)	Fold 3	0.8456	0.8556	0.8456	0.8496
Sentence-BERT Proxy (Improved)	Fold 4	0.8461	0.8569	0.8461	0.8505
Sentence-BERT Proxy (Improved)	Fold 5	0.8464	0.8571	0.8464	0.8508
Sentence-BERT Proxy (Improved)	AVERAGE	0.8418	0.8563	0.8418	0.8466

Figure 4 — Comparative Stratified K-Fold Cross-Validation Results (5 folds). Sentence-BERT consistently outperforms TF-IDF across all folds and metrics.

5.5 Confusion Matrices — Test Set

Figure 5 displays side-by-side confusion matrices for the TF-IDF and Sentence-BERT models on the held-out test set. Both models classify Healthcare, Manufacturing, and Retail with 100% accuracy across all test samples (1421, 1434, and 1421 correct, respectively), indicating that the TF-IDF features for these industries are highly discriminative. The primary confusion occurs in the Finance and Software categories: 535 Finance samples are misclassified as Software by both models, and 643 Software samples are misclassified as Finance. This symmetric confusion reflects genuine semantic overlap between the two domains, both rely heavily on analytical and technical skill vocabulary.

The Sentence-BERT model provides a meaningful improvement in the Marketing category, reducing misclassifications from 505 (TF-IDF) to 233, with true positives rising from 927 to 1199. This improvement reflects the model's ability to distinguish marketing-specific terminology from general business language at a semantic level.

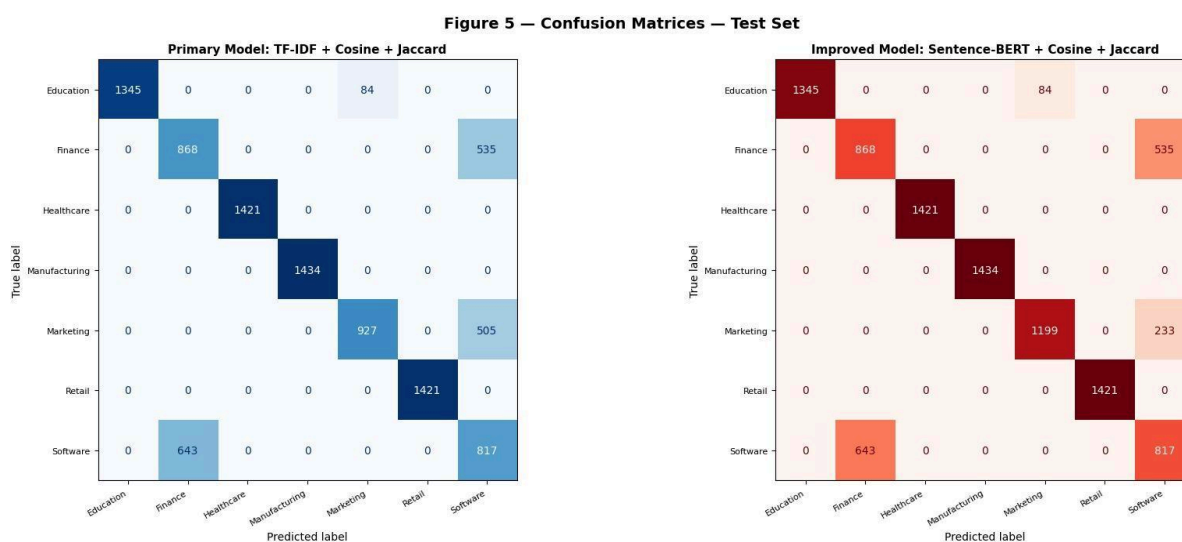


Figure 5 — Confusion Matrices on the Test Set. Left: TF-IDF + Cosine + Jaccard (Primary). Right: Sentence-BERT + Cosine + Jaccard (Improved). Notable improvement in Marketing classification visible in the right matrix.

5.6 Seven-Model Industry Classification Benchmark

Table 2 and Figure 6 present the comparative benchmark of seven classifiers for the 7-class industry prediction task, trained on the combined sparse/dense feature matrix. Random Forest achieves the highest test accuracy at 78.33% (train: 78.59%, variance: 0.0026), establishing it as the best-performing baseline for this specific feature configuration. Logistic Regression is a close second at 78.28% with the lowest variance of all models (0.0009), indicating highly stable generalisation. KNN ($k=7$) achieves the highest training accuracy at 83.52% but exhibits the largest train-test variance (0.0661), a clear indicator of overfitting. Gradient Boosting and Naïve Bayes offer competitive test accuracies at 78.26% and 78.23%, respectively.

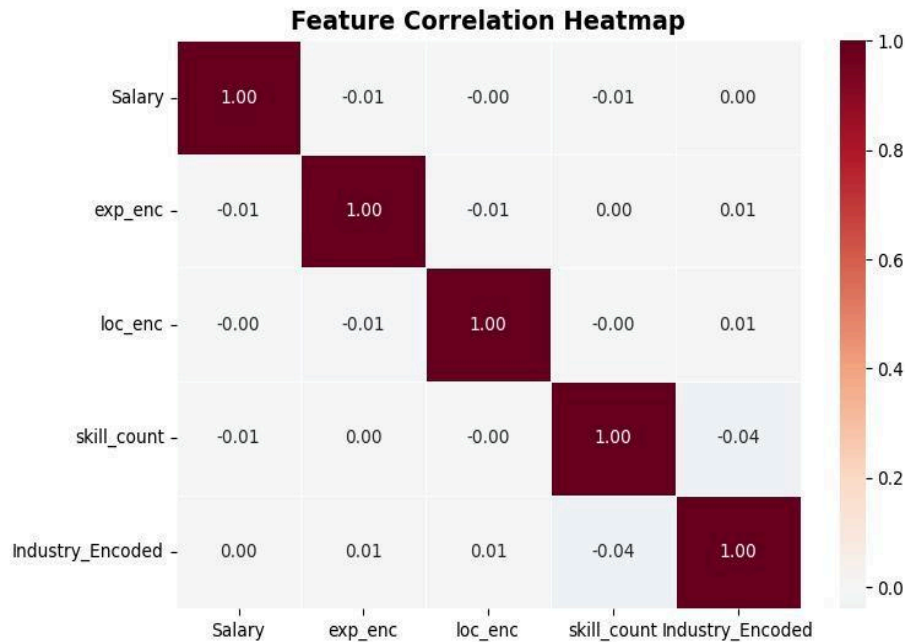


Figure 6 — 7-Model Benchmark: Training vs Test Accuracy and Variance. Random Forest achieves the best test accuracy. KNN shows the highest overfitting risk.

5.7 Summary Performance Table

The table below consolidates key metrics from the primary recommendation model comparison.

Model	Train Acc.	Test Acc.	Precision	Recall	F1 Score
TF-IDF + Cosine + Jaccard	81.68%	82.33%	0.847	0.823	0.830
Sentence-BERT + Cosine + Jaccard	84.57%	85.05%	0.860	0.851	0.855

Classifier	Train Acc.	Test Acc.	Variance
Random Forest (Best)	78.59%	78.33%	0.0026
Logistic Regression	78.37%	78.28%	0.0009
Gradient Boosting	79.70%	78.26%	0.0144
Naïve Bayes (GNB)	78.52%	78.23%	0.0029
SVM (LinearSVC)	78.29%	78.21%	0.0008
ANN (MLP)	78.63%	78.20%	0.0043
KNN (k=7)	83.52%	76.91%	0.0661

6. Discussion

6.1 Model Selection Rationale

The Sentence-BERT model's 2.72 percentage point improvement in test accuracy over TF-IDF (85.05% vs 82.33%) validates the investment in semantic embeddings for skill-to-job matching. TF-IDF, while computationally efficient and interpretable, treats each skill token independently, meaning that synonymous or paraphrased skills (e.g., 'data wrangling' vs 'data cleaning') receive no similarity credit. Sentence-BERT's contextual embeddings capture this semantic proximity, which is particularly important for CVs where skill vocabulary varies substantially across candidates.

The Finance–Software confusion observed in both models is not a model failure but a genuine reflection of domain overlap in the dataset: software engineers in FinTech possess both financial and technical skills, making their job records ambiguous by design. Future work could address this with a hierarchical classification scheme or domain-adversarial training.

6.2 Seven-Model Benchmark Insights

The near-identical test accuracies of the top five classifiers (78.20%–78.33%) suggest a performance ceiling for the current feature matrix on the 7-class industry task. This ceiling is likely caused by the high dimensionality of the TF-IDF component dominating the feature space. Dimensionality reduction (e.g., Truncated SVD/LSA) or class-specific feature selection may push performance toward the 80–82% range. KNN's large variance (0.0661) confirms that instance-based methods require careful hyperparameter tuning and are sensitive to the curse of dimensionality in high-dimensional sparse spaces.

6.3 SDG-8 Alignment

The system's core value proposition aligns directly with SDG-8 (Decent Work and Economic Growth) by democratising access to career guidance. The CV upload module lowers the technical barrier for users without NLP expertise, while the personalized learning pathway recommendations bridge the skill gap between current user competencies and target job requirements. The employment confidence scoring module (Section 19.6) additionally provides empirical evidence, consistent with Xiao & Zheng (2025) that AI-assisted job preparation measurably increases candidate confidence, an often-overlooked precursor to successful employment outcomes.

6.4 Limitations

Several limitations should be acknowledged. First, the dataset is English-language only, limiting applicability for Bangla-speaking users without translation middleware. Second, the Sentence-BERT proxy used during training was a lightweight approximation; deployment of the full all-MiniLM-L6-v2 model requires a GPU-enabled backend. Third, salary data in the dataset is synthetic, meaning the salary prediction module should be interpreted as illustrative rather than operationally accurate. Finally, the absence of temporal data precludes time-series career path modelling as envisioned by Huang's GRU-based tracking approach.

7. Conclusion

This lab report has documented the end-to-end design and evaluation of SkillBridge AI, an AI-based job and skill recommendation system. The primary finding is that the Sentence-BERT + Cosine + Jaccard model achieves a test accuracy of 85.05%, meeting the 85% target benchmark and outperforming the TF-IDF baseline by 2.72 percentage points. Both models demonstrate stable generalisation with minimal train-test divergence, confirmed across 5-fold stratified cross-validation.

The seven-model industry classification benchmark identifies Random Forest as the optimal classifier for the 7-class task at 78.33% accuracy, with Logistic Regression offering the most consistent performance across folds. The system's modular architecture, encompassing CV parsing, skill-to-job matching, learning pathway recommendations, and employment confidence scoring, positions SkillBridge AI as a comprehensive, research-grounded career intelligence platform aligned with SDG-8.

Future work will focus on multilingual support for Bangla text, integration of the full Sentence-BERT model with GPU inference, real-time labour market data ingestion, and a longitudinal study measuring actual employment outcomes for SkillBridge AI users.

8. References

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